Functional Consciousness: A Proxy Metric Using Self-Models

Frank W. Bergmann[0009−0005−1844−3269]

Private Researcher
fraber@fraber.de

Abstract. This paper proposes Functional Consciousness (FC) as a measurable architectural property: the observable capacity of a system to access and reason about internal representations of its own states. We introduce a computationally tractable metric based on self-models and their associated reasoning power, measured through informational richness and state-space expansion under inference. The resulting Functional Consciousness Score (FCS) is applied to benchmark systems with known internal structure, including a Waymo L4 autonomous vehicle. To extend the framework to black-box systems, we present Functional Self-Model Analysis (FSMA), an abductive methodology for inferring self-models from behavioral evidence. Applied to stream-of-consciousness literature, FSMA yields a catalog of self-models that serves as a reference for estimating functional consciousness in more complex biological and artificial agents. The resulting scores align with intuitive gradients of cognitive sophistication while remaining operationally grounded. Finally, we compare FC with major theories of consciousness and argue that several of their central functional claims become partially measurable within this framework.

Keywords: AI Benchmarking · AGI · Cognitive Architectures · Consciousness · IIT · Information Theory · Machine Consciousness · Philosophy of Mind

Introduction

An AI system cannot be genuinely "general" if it cannot become an object of its own reasoning. As systems move from narrow task execution toward autonomous, general behavior, the ability to represent, inspect, and reason about internal states becomes a central engineering problem. This capacity is required for AI safety, reasoning about an agent's own cognitive limits, and Theory of Mind (modelling others like yourself) [20]. It also addresses practical issues like memory reorganization and error reporting. However, it remains unclear how to define or measure this architectural feature.

This paper proposes "Functional Consciousness (FC)" to capture a system's observable capacity to access and reason about its internal states. We deliberately

sidestep the "hard problem" of consciousness [14] and "phenomenal consciousness"—the question of "what it feels like" to a subject.

Instead, we ground our work in functionalism [21] and access consciousness [6]. These frameworks define mental states by their functional roles in producing observable behavior. Viewed this way, Functional Consciousness becomes a tangible architectural property rather than an elusive quality.

Definition 1 (Functional Consciousness, FC). The observable capacity of a system to access and process internal representations of its own states to produce behavior.

By adding the "self-model" as a unit of analysis, we can apply the tools of information theory [5] to measure the predictive and reasoning power of these models. In this section, we validate this "white-box" metric by benchmarking agents with known structures, including a Waymo L4 autonomous taxi.

We then extend this method to agents with unknown internal structures. While this involves coarser approximations, we can still provide numerical estimates for their FC. We validate this "black-box" approach by benchmarking a range of agents. Finally, we discuss how FC instantiates several well-known theories of general consciousness and inherits their metaphysical claims.

Self-Models

The term "self-model" originates from Thomas Metzinger's Self-Model Theory of Subjectivity (SMT) [17], where it refers to a single, unified model. We propose a more granular approach: treating consciousness as an aggregate of multiple domain-specific self-models (e.g., kinematic, social, task-oriented). This echoes Michael Graziano's Attention Schema Theory [13] and Yann LeCun's work on World Models [16]. The concept was informally proposed at AGI-2015 [2]; here, we provide a formal grounding:

Definition 2 (Self-Model). A self-model is any internal representation that functionally models the system's own states, processes, or capacities, and whose content is available to global reasoning processes.

Consider the following statement as an intuitive indicator:

> "I can run faster than you."

This phrase implies the agent possesses an internal representation of its physical capabilities. To produce such a comparison, the underlying model must exhibit several observable characteristics:

- **Domain:** The functional area covered (e.g., kinematics).
- **Conceptual Breadth (B):** The range of features tracked (e.g., "running" and "jumping" performance).
- **Conceptual Depth (D):** The granularity of information (e.g., statistics on peak velocity or endurance).

– **Reasoning Power (P):** The performance of the inference engine operating on the model (e.g., extrapolating performance to a future race).

Quantifying these properties can be difficult. Domains often overlap; a "spatial" model may be inseparable from a "body" model in a vacuuming robot. Furthermore, a spatial self-model may be hard to distinguish from a first-order model of the 3D environment.

Note the absence of "interconnection" between self-models as proposed by Integrated Information Theory [24]. Instead, self-models are "illuminated" by attention, making their contents available to global reasoning processes which effectively perform the integration.

Scoring Self-Models

Our scoring system rewards agents that possess detailed, broad, and— most importantly—utilizable self-representations. A self-model is considered "better" if it covers a wide range of phenomena (Conceptual Breadth), provides high detail (Conceptual Depth), and allows the system to draw actionable conclusions (Reasoning Power).

We define the Functional Consciousness Score (FCS) of a single self-model as the product of its representational capacity and its reasoning power. This multiplicative structure explains why legacy AI systems often exhibited low FC despite possessing detailed data: their reasoning engines were too weak to extract significant utility from the model. Conversely, a powerful reasoning engine (like a stateless LLM) scores zero if it lacks a persistent self-model to operate upon.

To formalize this, we draw on the language of predictive information [5] to express representational capacity (R) and reasoning power (P).

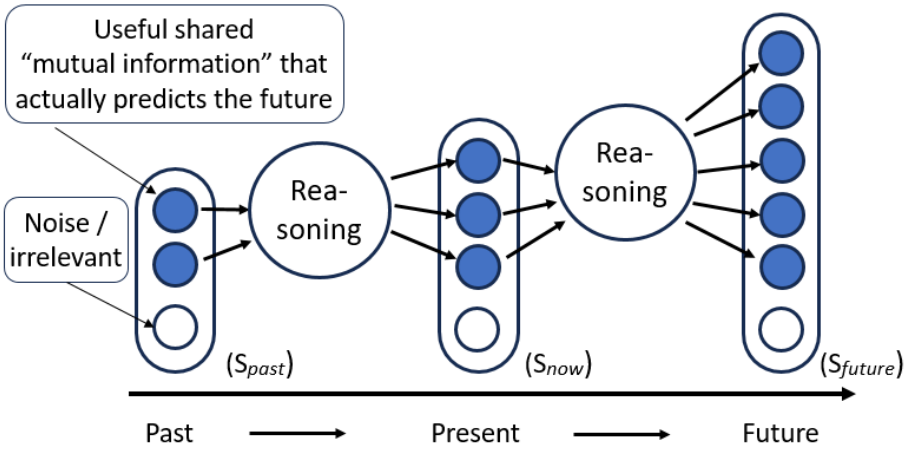

Fig. 1. Scheme of State-Space Expansion and Mutual Information.

Bialek et al. argue that most information a system processes is actually useless noise. True complexity lies in the **predictive information**—the subset of mutual information from the past that actually helps predict the future.

Definition 3 (Functional Consciousness Score, FCS). For a self-model m in domain f, the Functional Consciousness Score is defined as:

$$FCS(m) = R(m) \cdot P(m) \quad (1)$$

where:

- $R(m)$ is the **Representational Capacity**, the sum of mutual information $I(m_i; s_i)$ between each independent self-model variable m_i and the corresponding system state variable s_i.
- $P(m)$ is the **Reasoning Power**, measured as the **state-space expansion rate**—the rate at which distinguishable, reachable states grow through inference over time.

This multiplicative structure highlights the "Map vs. Agent" distinction. A detailed map (high R, zero P) is not functionally conscious. A powerful LLM with no ground truth (zero R, high P) is equally stagnant.

Representational capacity $R(m)$ can be decomposed into two factors:

$$R(m) = B(m) \cdot \bar{D}(m) \quad (2)$$

where:

- $B(m)$ **(Breadth)** is the effective dimensionality: the number of variables tracked above a noise threshold ϵ.
- $\bar{D}(m)$ **(Depth)** is the average mutual information per tracked variable: $R(m)/B(m)$.

Combining Multiple Self-Models

An agent typically consists of multiple self-models. To score the entire system, we must consider how global reasoning integrates these disparate representations.

No cross-reasoning If there is no reasoning across models, the total score is simply the sum of the individual scores: $FCS_{agent} = \sum_j FCS(m_j)$.

Perfect cross-reasoning At the other extreme, an agent capable of drawing all possible conclusions across its self-models achieves "informational closure." In this case, the total capacity is additive ($R_{agent} = \sum_j R(m_j)$), but the reasoning power becomes exponential with the number of self-models: $P_{agent} = \prod_j P(m_j)$.

This aligns with the highest levels of Φ in Integrated Information Theory [24], where the system exists as a single, irreducible causal entity. The product structure

provides a tractable way to measure **Integration Density**, aligning with the mechanics of a perfect global workspace [1, 12].

Practical cross-reasoning Real-world agents vary in how many state variables are accessible to global reasoning. Modern LLMs outperform legacy symbolic systems by several orders of magnitude in this cross-domain integration. Techniques like abstraction and efficient heuristics expand the accessible state-space and thus directly increase the FCS_{agent}.

Functional Consciousness Score for Waymo L4 Spatial/Kinematic

We illustrate the FCS metric by calculating the score of the spatial/kinetic self-model of the "white-box" Waymo L4 autonomous taxi [27].

Representational Capacity We estimate $R_{waymo} \approx 480$ bits by multiplying the ~40 self-model state variables (B_{waymo}) by an average depth (D_{waymo}) of ~12 bits per variable. The variables include:

- **Kinematic Self-Model:** 12 variables (Position x, y, z; Velocity v_x, v_y, v_z; Acceleration a_x, a_y, a_z; Angular rates $\omega_{pitch,yaw,roll}$).
- **Actuator Self-Model:** 8 variables (Steering torque, braking pressure, wheel rotational speeds).
- **Task/Trajectory Self-Model:** ~20 variables (Active path spline parameters, distance-to-lane-center, time-to-arrival, collision buffers).

We estimate an average mutual information $I(m_i; s_i) \approx 12$ per variable with 14 bits effective for floating-point values and 1–4 bit for discrete state variables.

Reasoning Power We calculate P_{waymo} as the state-space expansion rate—the growth of the "conclusion manifold" during inference. Waymo uses Model Predictive Control (MPC) and Monte Carlo simulations for trajectory planning.

- **Initial State ($t = 0$):** 480 bits.
- **Horizon ($t = 5s$):** The system simulates thousands of branched futures based on candidate actions and actor interactions.
- **Expansion Factor:** If the inference engine generates a probability distribution over 2^{10} distinguishable and reachable future states, the expansion rate is the bit-growth of this set.

Following Bialek et al. [5], for a $K = 40$ parameter model with $N \approx 100$ inference steps per cycle:

$$P \approx \frac{40}{2} \log_2(100) \approx 20 \times 6.64 \approx 133 \text{ expansion units.}$$

As a result, we get:

$$FCS_{waymo} = R \cdot P = 480 \times 133 \approx \mathbf{64,000} \text{ FC Points} \quad (3)$$

This score indicates that while a Waymo taxi has a detailed model of its physical self and future trajectories (R), its functional consciousness is limited by narrow reasoning in its specific domain. It lacks the cross-domain "reasoning manifold" expansion seen in biological agents.

Self-Model Evaluation Comparison Table

We evaluate selected self-models of several contemporary systems against the human baseline. We have intentionally chosen a qualitative scale because we have to mix quantitative and qualitative values. Details for each system are available in the companion repository [4].

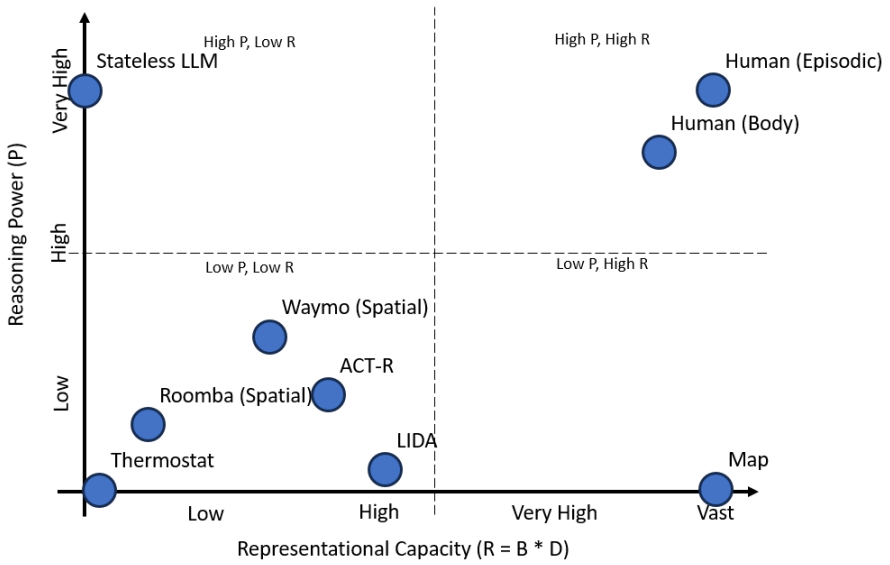

Fig. 2. Reasoning Power vs Representational Capacity (from Table 1).

Table 1. System Comparison Table

System	B (vars)	$\bar{D}$ (bits/var)	P	FCS	Details
Stateless LLM	0	0	Very High	0	
Map	Vast	Rich	0	0	
Roomba (kinematic)	~18	~8	~39	~5.6k	roomba.md
Waymo (kinematic)	~40	~12	~133	~64k	waymo.md
LIDA (?)	~20	~4	Low	Small	lida.md
ACT-R (?)	~20	~8	Moderate	Medium	act-r.md
Human (kinematic)	~550	~10	High	~10M	
Human (episodic)	Vast	Rich	Very High	Very Large	

While some AI systems match human performance in narrow domains (e.g., an LLM's raw reasoning power), they rarely combine high capacity and reasoning within a single self-model. Furthermore, the human mind integrates many such models across domains, leading to an exponential growth with the number of models.

Functional Self-Model Analysis (FSMA)

The FCS presented in previous sections provides a structural estimate for "white-box" agents. However, many systems are black boxes, accessible only through their behavior. For these cases, we introduce Functional Self-Model Analysis, a behavioral inference method that detects the presence and richness of self-models through the predictive structure of observed outputs.

The intuition is straightforward:

> "If an agent can repeatedly and consistently talk about a state, it must possess a functional model of that state."

This is analogous to functional data dependency [8] and abductive reasoning—inferring the necessary preconditions for an observed outcome.

The VAT Intuition To illustrate this, consider an automated accounting system that calculates Value Added Tax (VAT) for international transactions. By law, the VAT rate depends on:

- The origin of the item.
- The destination of the item.
- The category of the item (e.g., food vs. services).

If the system consistently identifies the correct VAT across diverse invoices, we can abductively conclude that:

1. The system possesses data representing the origin, destination, and category.
2. The system possesses a functional mapping VAT $= f$(origin, destination, category).

Whether this mapping is a lookup table, a hard-coded rule, or a latent neural vector is irrelevant to the functional analysis; its presence is a prerequisite for the observed behavior.

Definition 4 (Functional Self-Model Analysis, FSMA). FSMA identifies the minimal set of self-models $M(f)$ required for a consistently produced observable output f:

$$M(f) = \{m_i \mid m_i \text{ is a minimal self-model required for } f\} \quad (4)$$

Each self-model m_i is defined as a tuple $\langle D_i, S_i, O_i \rangle$, where:

- D_i **(Data Domains):** The specific areas represented (e.g., spatial, temporal).

- S_i **(Structural Form):** The architecture of the representation (e.g., 3D scene graph, directed acyclic graph, attention map).
- O_i **(Operations):** The functional capabilities enabled (e.g., simulation, prediction, counterfactual reflection).

"Minimal" refers to the model with the lowest representational capacity required to satisfy the observation. Identifying the absolute minimum may not always be possible for practical purposes, so this criterion helps decide between multiple candidate solutions.

Empirical Evaluation of FSMA

To test the FSMA methodology, we required datasets rich in references to internal states. We initially surveyed several "Descriptive Experience Sampling" (DES) collections [15], which provide systematic reports of human inner experience. However, many existing DES datasets are either coded with restrictive categories or focus on meta-level interviews rather than raw streams of thought.

We broadened our search to early 20th-century stream-of-consciousness literature. While literary texts are crafted for aesthetic effect, they offer a high-density representation of subjective mental flows. The internal logic of FSMA still holds against the "parroting" or "word-to-world" fallacies: to repeatedly and consistently describe thoughts and perceptions as these authors do, the narrator must possess sophisticated underlying self-models. For our primary evaluation, we selected Virginia Woolf's "The Mark on the Wall" [28] for its manageable length and extraordinary richness in meta-cognitive reflection. The annotated text is available in the companion repository [4].

The Self-Model Catalog During our FSMA of "The Mark on the Wall", we extracted all self-referential expressions to identify candidates for the "minimal" self-models required to generate them. This process yielded a comprehensive Self-Model Catalog.

Through this fieldwork, we encountered the potential fallacies of Schleiermacher's hermeneutic circle [24], which we navigated using two distinct modes of analysis:

1. **Bottom-Up Analysis:** An inductive process extracting self-model candidates directly from the text, grouped by shared data requirements.
2. **Top-Down Analysis:** A deductive validation of existing theoretical frameworks (e.g., a specific cognitive architecture) against the models identified in the text.

Bottom-up analysis reveals the inherent complexity of self- representation. For example, we initially identified overlapping models like a "dual self" (private vs. public) and a "multi-faceted self" (context-dependent identity). While philosophically interesting, these often function as special cases of broader "Theory of

Mind" (ToM) or "Narrative Self" models. To maintain a lean, benchmark-ready catalog, we aggregate these into general categories based on functional similarity.

In contrast, top-down analysis carries a significant methodological risk: if the analysis begins with a predefined architecture, the investigator may primarily detect only the models that the framework already assumes. This can lead to confirmation bias.

We address this dilemma by decoupling the FSMA methodology from any single catalog. This allows researchers to apply their own theoretical frameworks, leaving scientific discourse to determine which catalogs offer the most predictive power for AGI.

The SBR-Catalog of Self-Models The following catalog results from a bottom-up FSMA of Woolf's text, filtered through the lens of the Scene-Based Reasoning (SBR) cognitive architecture [3]. SBR details several first-order subsystems and assumes an attention subsystem [13] that can focus on both external objects and internal self-models.

The SBR-derived catalog consists of 46 self-models across ten functional areas (see the companion repository [4]):

- Body (5): `body-3d`, `body-kinematic`, `body-sensor`, `body-actuator`, `body-energy`.
- Spatial (4): `spat-relative`, `spat-trajectory`, `spat-collision`, `spat-tool`.
- Action/Planning (5): `action-tree`, `action-perform`, `action-progress`, `action-plan`, `action-improv`.
- Goal/Motivation (3): `goal-tree`, `goal-reward`, `goal-conflict`.
- Cognitive (5): `episody`, `episody-narrative`, `episody-time`, `mem-avail`, `learn-rate`.
- Informational/Knowledge (7): `inf-know`, `inf-fresh`, `inf-creative`, `inf-consistency`, `inf-reasoning`, `inf-hypo`, `inf-confidence`.
- Emotional/Affective (4): `mood`, `mood-needs`, `mood-stress`, `mood-load`.
- Social/Interaction (6): `social-tom`, `social-role`, `social-comm-state`, `social-trust`, `social-influence`, `social-empathy`.
- Meta/Reflexive (4): `meta-attention`, `meta-self-awareness`, `meta-explain`, `meta-accuracy`.
- Ethics/Safety (3): `ethics`, `ethics-safety`, `ethics-drift`.

Perception is excluded from this catalog as a first-order activity with several self-models monitoring the perceptual process (e.g., `body-sensor` for sensory limits). "Awareness" of perception is thus modeled as attention toward these self-models.

Benchmarking Agents with the SBR Self-Model Catalog

While previous sections focused on the numerical FCS of "white-box" agents with limited capabilities, here we benchmark biological and more complex AI systems.

Methodology of Comparison Estimating a Functional Conscousness Score for restricted human self- models, such as `meta-attention`, may be possible using methods similar to our Waymo analysis. However, calculating scores for larger models like the narrative self or social interaction is currently beyond the scope of this work.

For this evaluation, we use a simplified approximation of the FCS. In this model, we assign $B = 1$ per self-model domain and restrict $\bar{D} = 1$ and $P \in \{0, 0.01, 1.0\}$ for no reasoning, legacy symbolic reasoning, and LLM/human-level reasoning. This "1.5- bit" model assumes a similar "size" for all self-models and uses an arbitrary approximation of reasoning power to facilitate rapid benchmarking.

We justify this coarse approach by the exponential growth of the reasoning manifold ($P_{agent} = \prod_j P(m_j)$) that dominates the total score. Future work will operationalize higher-fidelity measures using mutual information and state-space expansion.

We benchmarked the same systems as in Table 1, aggregating the FCS score per functional area of the catalog, and dividing by the maximum possible score per area in order to get a percentage value.

The following systems were selected for benchmarking:

- **Roomba:** These robots use SLAM [11] to build spatial representations. As analyzed earlier, they possess basic spatial self-models but have very limited reasoning power (P).
- **Waymo L4:** The Waymo possesses sophisticated spatial self- models with integrated uncertainty, health monitoring, and trajectory simulation. It exhibits cross-domain reasoning (e.g., correlating sensor reliability with trajectory planning).
- **LIDA Cognitive Architecture:** LIDA is designed to model the "conscious cycle" via Global Workspace Theory [12]. While it possesses numerous self-models, its symbolic reasoning engine is limited compared to modern neural approaches, leading to high R but low P.
- **Average Human Baseline:** A human agent possesses the full catalog of self-models, all available to global reasoning. This represents our normalization baseline (score 1.0).

(Visualizations comparing these systems are provided in the companion repository [4].)

Relation with Other Theories of Consciousness

Functional Consciousness (FC) was not proposed as a competing metaphysical account of phenomenal experience or the "hard problem." However, FC instantiates and details several major theories, inheriting their claims and serving as a bridge between abstract philosophy and concrete engineering.

Integrated Information Theory Giulio Tononi's Integrated Information Theory (IIT) [24] is one of the most prominent theories, identifying consciousness with integrated information (Φ) within a causal system. However, it is often criticized for computational intractability and a perceived lack of empirical falsifiability.

We can ground these ideas by mapping them onto measurable functional metrics: IIT "differentiation" corresponds to the representational capacity of self-models, while "integration" corresponds to reasoning power.

It is even possible to calculate the Φ of an agent where subsystems are governed by multiple, overlapping self-models. Total integration can be viewed as the difference between collective reasoning power and the sum of the reasoning power of partitioned modules. We can approximate this by measuring how much reasoning power (P) is lost when we disconnect self-models from their underlying subsystems:

$$\Phi_{FCS} = P(S) - \sum_j P(module_j) \quad (5)$$

Where:

- $P(S)$: The total reasoning power of the entire integrated system.
- $module_j$: The j-th partitioned subsystem combined with its dedicated, non-shared self-models.

In summary, FC decomposes Φ into measurable components and operationalizes "interconnection" using a standard apparatus for reasoning and attention [13].

Higher-Order Theories Higher-Order Theory (HOT) [22] and Higher-Order Perception (HOP) posit that a mental state is conscious only when it is the object of a higher-order state. Functional Consciousness naturally instantiates this: its multi-level architecture use self-models (m_i) to represent and reason about first-order states (s_i).

In this framework, "higher-order" is operationalized as the attention mechanism focusing on self-models. A state s_i becomes an object of access consciousness when the system's reasoning engine (P) processes an associated self-model (m_i). Recursion—such as the `meta-attention` model monitoring attention itself—creates the **meta- reflective cascades** characteristic of advanced cognition.

Workspace and Predictive Theories Global Workspace Theory (GWT) [1] suggests that consciousness arises when information is "broadcast" to a wider network of cognitive processes. FC formalizes this broadcasting as the integration of multiple self-models into a unified global reasoning space. The product structure of reasoning power ($P_{agent} = \prod_j P(m_j)$) provides a direct measure of this workspace's density.

Predictive Processing (PP) [7] describes the brain as a prediction engine. Our use of predictive information [5] to measure representational capacity (R) grounds FC firmly in the PP tradition. In this view, consciousness is the process of minimizing the prediction error of the agent's own internal self-models.

Summary Position Many leading theories of consciousness implicitly rely on some form of self-representation, global integration, or meta-reflection. Functional Consciousness makes these structural properties explicit and quantifiable. In this sense, FC serves as an implementation layer that instantiates and unites these diverse theoretical accounts.

Conclusions

Functional Consciousness (FC) offers a computationally tractable behavioral metric for measuring how an agent models and reasons about its own internal states. By moving beyond the "hard problem," we provide a framework that is useful for both cognitive science and AI engineering.

Our results demonstrate that FC can successfully benchmark a diverse range of systems—from autonomous taxis to 20th-century literature. We show that:

1. **Metric Clarity:** The $FCS = R \cdot P$ formula captures the essential trade-off between representational capacity and reasonive utility.
2. **Integration Value:** The exponential growth of cross-domain reasoning explains the significant leap from narrow AI to general biological consciousness.
3. **Benchmarking Utility:** FSMA provides a systematic way to extract and compare self-models from behavioral output, even in "black-box" systems.

Outlook for AGI As we move toward Artificial General Intelligence (AGI), FC provides a vital safety and evaluation tool. An agent that cannot accurately reason about its own capabilities, goals, and limitations is inherently unreliable. By measuring and optimizing for Functional Consciousness, we can build systems that are not only more intelligent but also more transparent and self-aware.

To increase FC in future systems, we recommend building AI with **inspectability** in every subsystem and exposing the results to global reasoning via an attention filter. Key missing capabilities in current AI include predictive world simulation and periodic self- checks to increase abstraction.

Future work will expand the SBR-Catalog and develop higher-fidelity instruments for measuring information-theoretic "bits" in complex neural architectures. The goal is to move from qualitative approximations toward a rigorous, universal scale for machine consciousness.

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ToDo

- Remove the self-model names from Table 1 and Fig 1
- Add section on LLMs with text self-models
- Review the evaluation of the various AI agents and their position in the 2x2
- Add an excerpt from the SBR self-model catalog?
- Separate and beautify the evaluations of the various AI agents and prepare them as one attachment.
- Roomba | ~4 | ~10 | Low | Small | roomba.pdf |
- Waymo | ~50 | ~12 | ~165 | ~100k | waymo.pdf |
- LIDA (Episodic) | ~20 | ~4 | Low | Small | lida.pdf |
- ACT-R (Episodic) | ~20 | ~8 | Moderate | Medium | act-r.pdf |
- Human body eval (done)
- Human episodic eval

Fig.2: -* Human Baseline: wo kommt die denn her? Woher weißt du denn, wie viele State Variables wir im Hirn haben (oder ob wir überhaupt welche haben)? -* ACT-R: ich bezweifle, wie du einer Architektur überhaupt einen Consciousness-Wert zuordnen kannst. Das kommt ja immer noch drauf an, wie sie für je weilige Anwendungen instanziiert wird. Du kannst ja auch nicht aufgrund einer Programmiersprache sagen, dass jedes Programm eine bestimmte Effizienz a ufweisen wird.

- The “minimal” self-model from FSMA changes, as soon as new output appears that can’t be explained with the last version of a self-model. This is a necessary effect.
- Radar charts are just a different way to visualize the FCS. It’s focus is on the distribution of self-models across the different areas, allowing to intuitively judge the type of agent. However, it does not add new information. Please note that the scale of the chart is implicitly logaritmic, because the FC score includes the P factor that is the product of the reasoning powers of the participating self-models, assuming a certain degree of “integration density”.
- rename GiHub “open-conscousness” to “functional-conscousness”